A dimensional risk assessment of Generative AI in SSCI assessments

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In collaboration with the SSCI GenAI Working group

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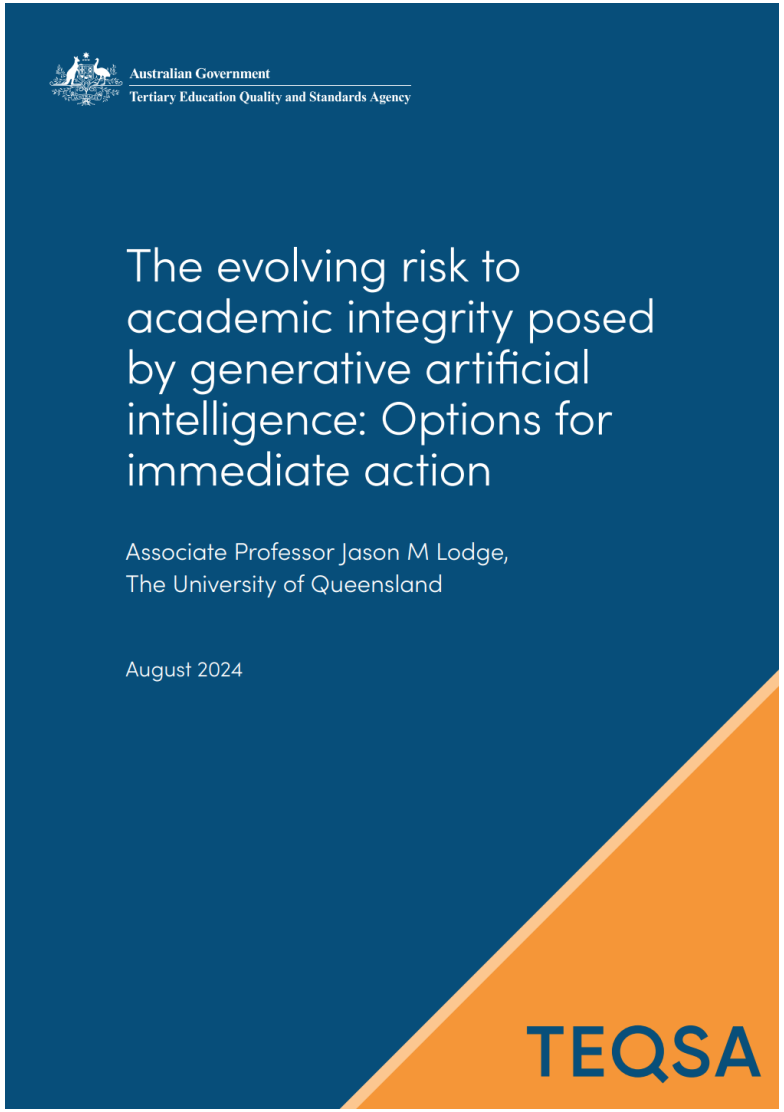
AI is a rapidly emerging and changing tool

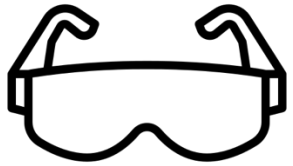


- Widespread concerns that AI-assisted cheating may undermine learning and assessment
- Estimating scale of use is difficult because it is undetectable
- Innappropriate use of AI is an immediate risk to academic integrity
- This risk is compounded by the pace at which these tools are developed

https://www.teqsa.gov.au/sites/default/files/2024-08/evolving-risk-to-academic-integrity-posed-by-generative-artificial-intelligence.pdf

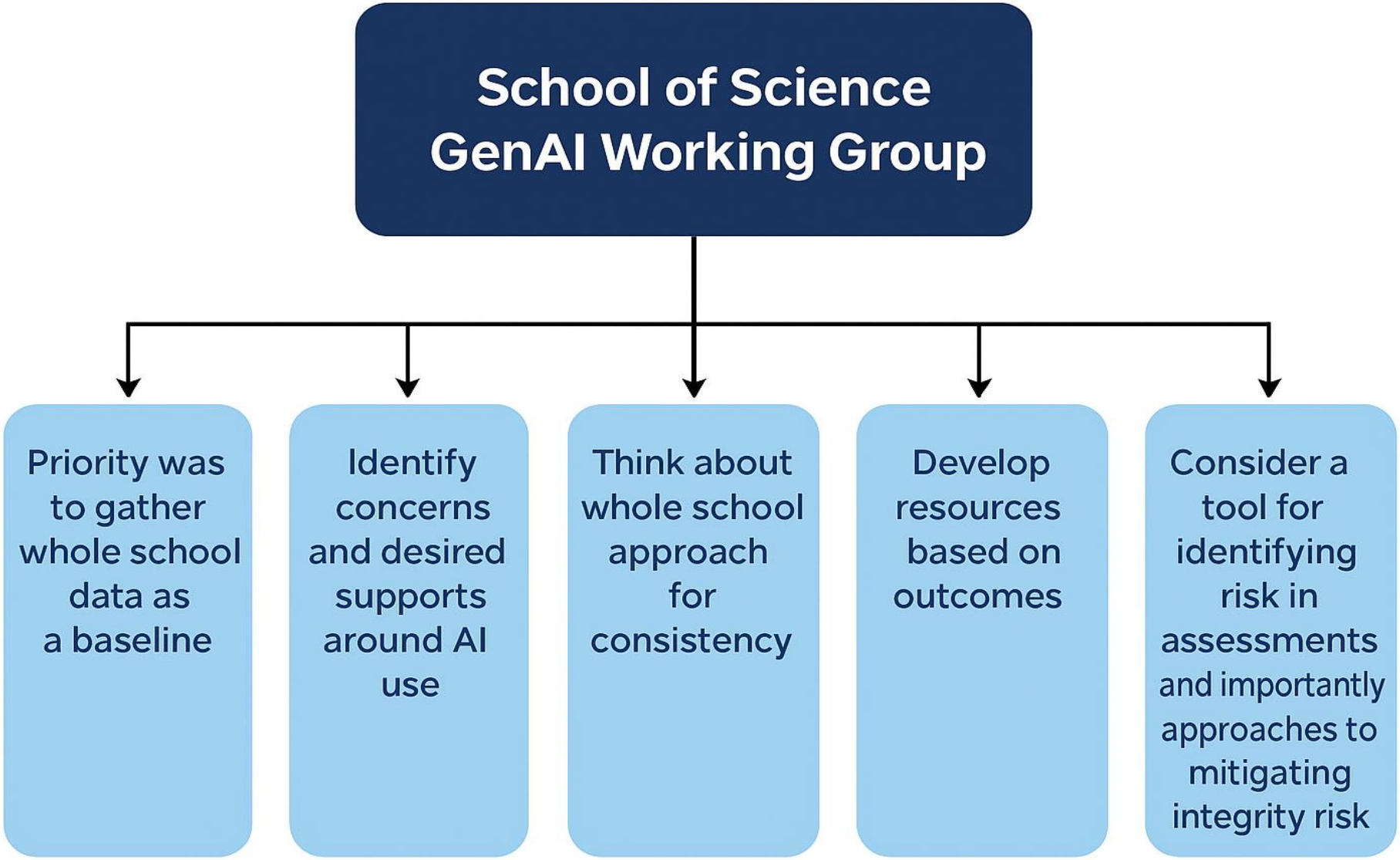
New tools are risky! But that doesn't mean we ban them

- In labs, new reagents or equipment don't mean we stop experiments
- Any tool used incorrectly is risky
- We use **risk assessments**, **PPE**, **protocols**, **controls**



GenAI is the same: a novel tool, not an existential threat

Making a risk assessment for SSCI – The Team



“What is the *actual* risk level of GenAI across assessments in the School of Science, and how can we measure it consistently?”

- Traditional risk assessments focus on assessment *type* alone
- But factors such as *authenticity*, *personalisation*, and *guidance* also shape AI vulnerability
- Needed: a scalable, transparent, multi-dimensional framework

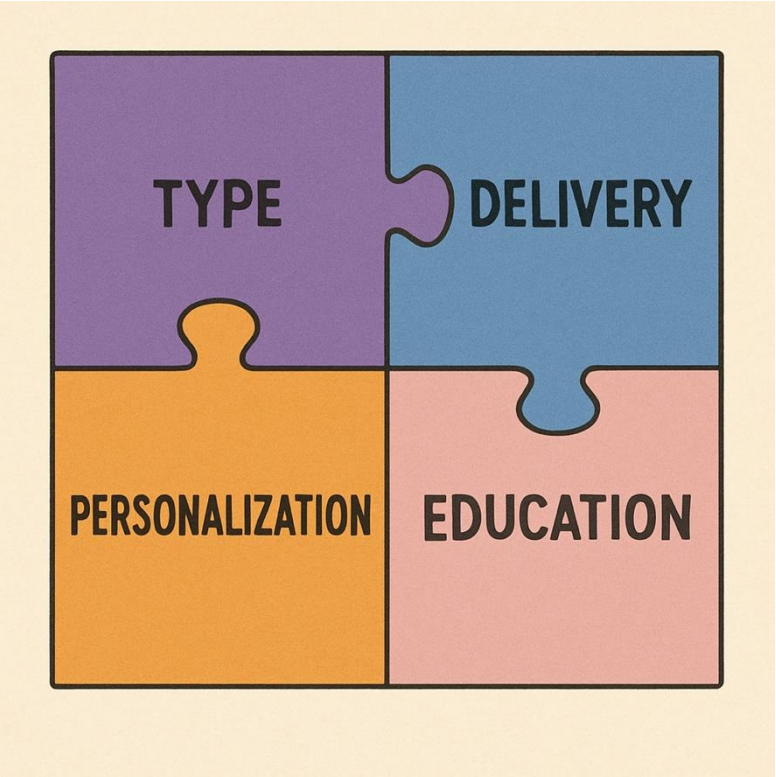
The four dimensions Risk Matrix

A simple tool to indicate potential risk to GenAI use in assessments

Data sources:

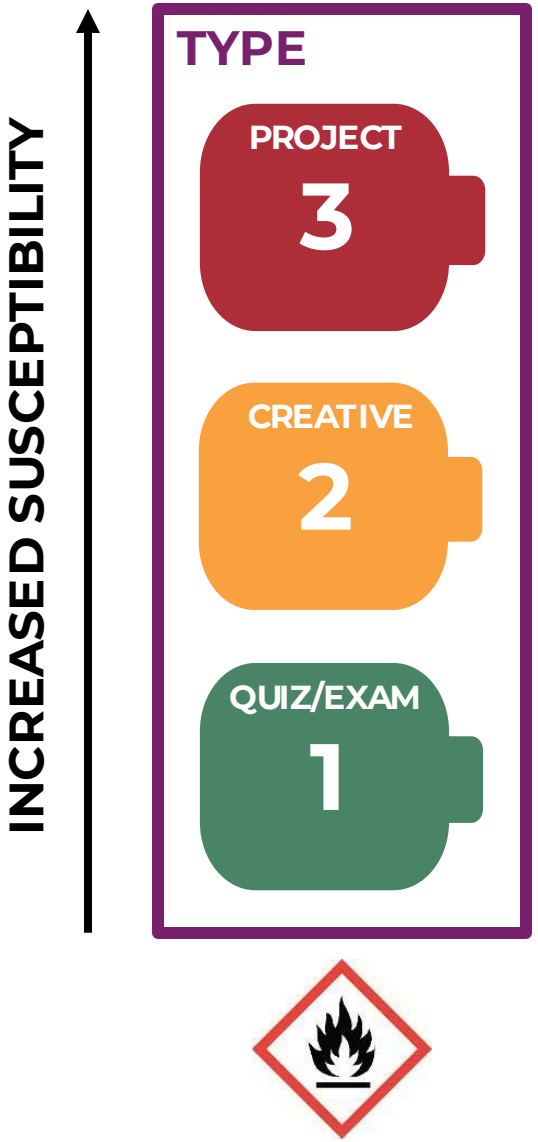
- Subject outlines
- Interviews with Subject Coordinators
- **48 undergraduate subjects**
- **All assessments** per subject **(big data)**

- GenAI related to integrity of assessment and assurance of learning
- The risk matrix score is determined on 4 dimensions and considers assessment weighting
- Scores for each assessment and/or average for subject



How did we turn these dimensions into a score?

Risk dimension: Assessment type



Extended written responses or data-driven investigations where students synthesise sources or produce original analysis → *literature review, lab report, design project*

Outputs requiring verbal communication, visual design, or creative interpretation → *oral presentation, video submission, poster, or artwork*

Structured assessments testing recall, problem-solving, or application under timed or invigilated conditions → *final/midsession quizzes*

Risk dimension: Delivery mode

Fully online submission or completion with no invigilation, enabling unrestricted access to generative AI tools

Online submission or digital exam with some monitoring, or assessment with a mixed mode of submission

Conducted on campus without digital tools and/or under invigilated conditions, limiting AI use

Mitigation dimension: Authenticity/personalisation

INCREASED SUSCEPTIBILITY

TYPE	DELIVERY	AUTHENTICITY
PROJECT 3	ONLINE 3	NOTHING 0
CREATIVE 2	HYBRID 2	PERSONAL -1
QUIZ/EXAM 1	IN PERSON 1	BODY OF WORK -2

Applies to any student, no personalisation

Some unique aspect that cannot be easily generated/analysed with AI → *shared dataset or case study, lab experiment, personalised dataset, placement or fieldwork*

Unique progression of student's learning observed frequently during creation

Mitigation dimension: Guidance

INCREASED SUSCEPTIBILITY

TYPE	DELIVERY	AUTHENTICITY	GUIDANCE
PROJECT 3	ONLINE 3	NOTHING 0	POLICY 0
CREATIVE 2	HYBRID 2	PERSONAL -1	GUIDELINE -1
QUIZ/EXAM 1	IN PERSON 1	BODY OF WORK -2	EDUCATION -2

Institutional AI policy in subject outline

Explicit task-level guidance provided

Explicit in class discussion, examples, or resources educating students about appropriate AI use

Risk calculation

INCREASED SUSCEPTIBILITY

TYPE

- PROJECT 3
- CREATIVE 2
- QUIZ/EXAM 1

DELIVERY

- ONLINE 3
- HYBRID 2
- IN PERSON 1

AUTHENTICITY

- NOTHING 0
- PERSONAL -1
- BODY OF WORK -2

GUIDANCE

- POLICY 0
- GUIDELINE -1
- EDUCATION -2

× weighting = RISK SCORE

Examples

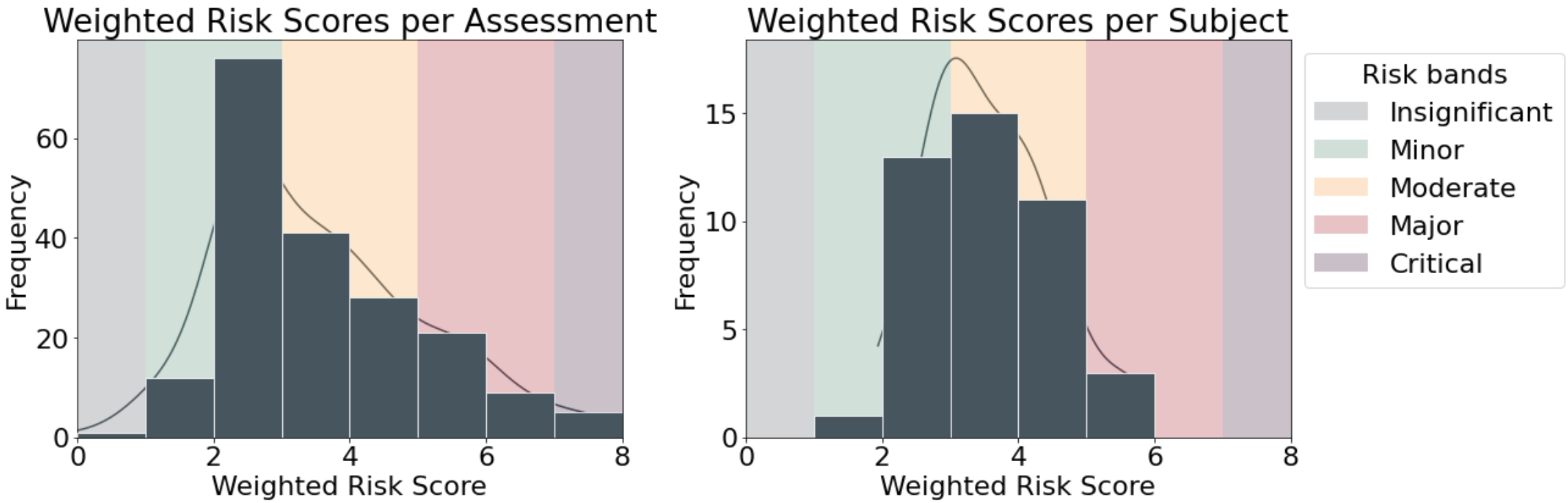
Project completed online without personalisation or education (35%)

PROJECT	ONLINE	NOTHING	POLICY			
3	3	0	0	× 1.35	=	8.1
						MAJOR

Final exam invigilated on campus (45%)

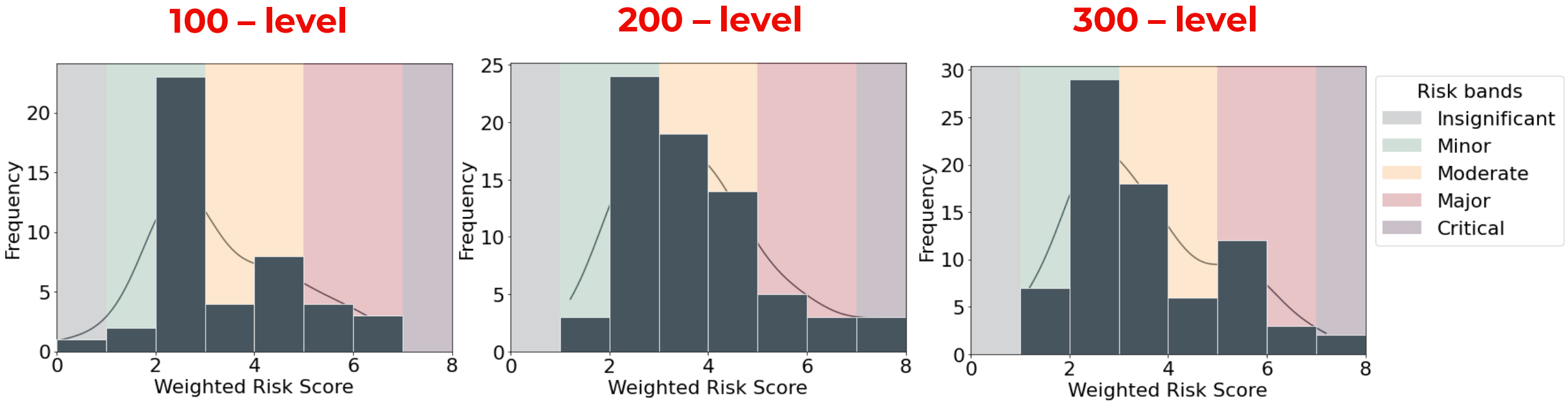
QUIZ/EXAM	IN PERSON	NOTHING	POLICY			
1	1	0	0	× 1.45	=	2.9
						MINOR

Overall risk distribution

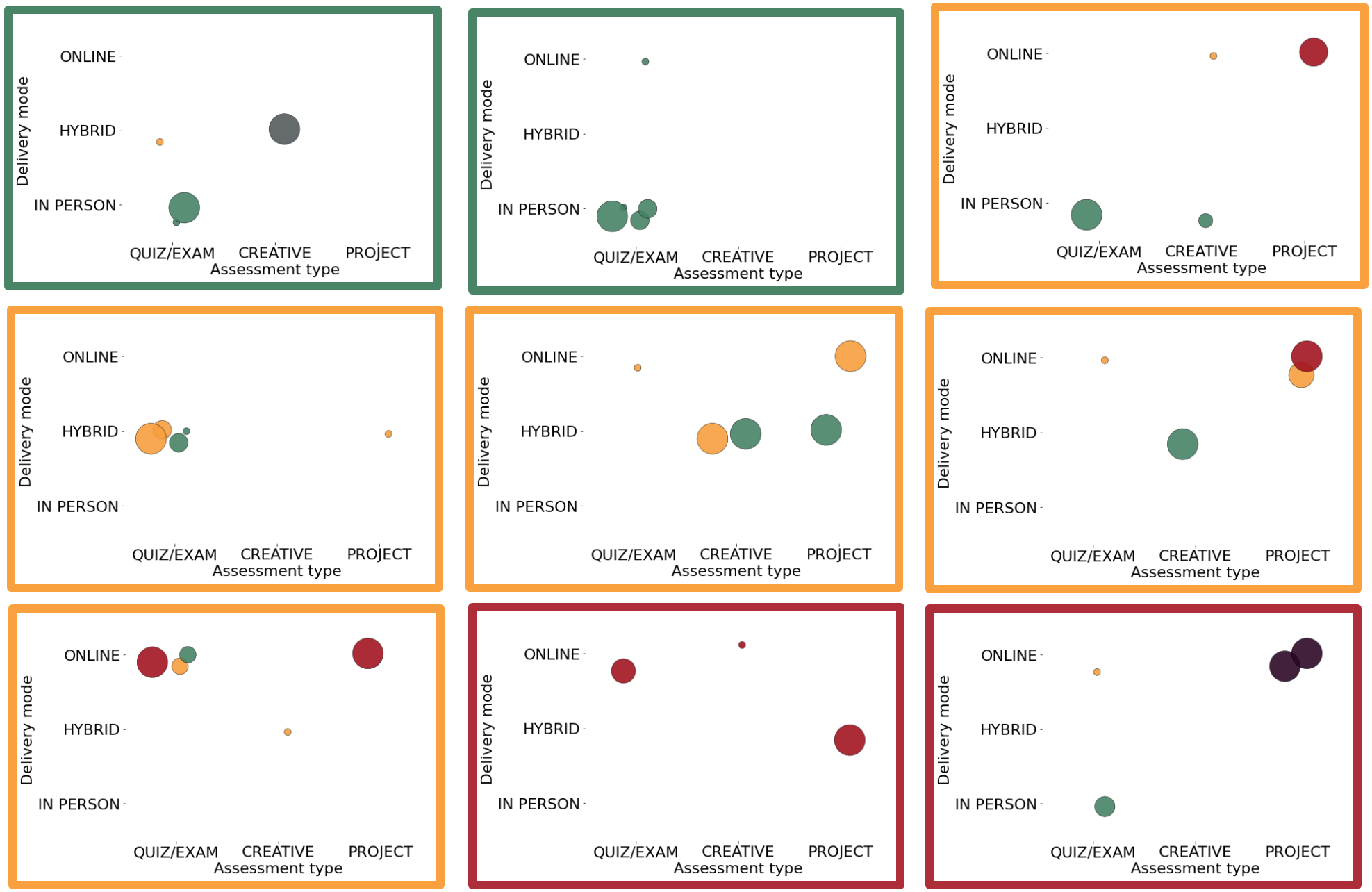


💡*Maximum score for a 100% assessment would be 12*

Risk distribution by level



Risk distributions per subject



Assessment Weighting

Risk rating

- Insignificant
- Minor
- Moderate
- Major
- Critical

Before and after PPE

What can I do to reduce risk?

- Personaliased data
- On-campus partial invigilation (if appropriate)
- Explicit AI guidance at task level
- Reduce weightings

… and many more options!

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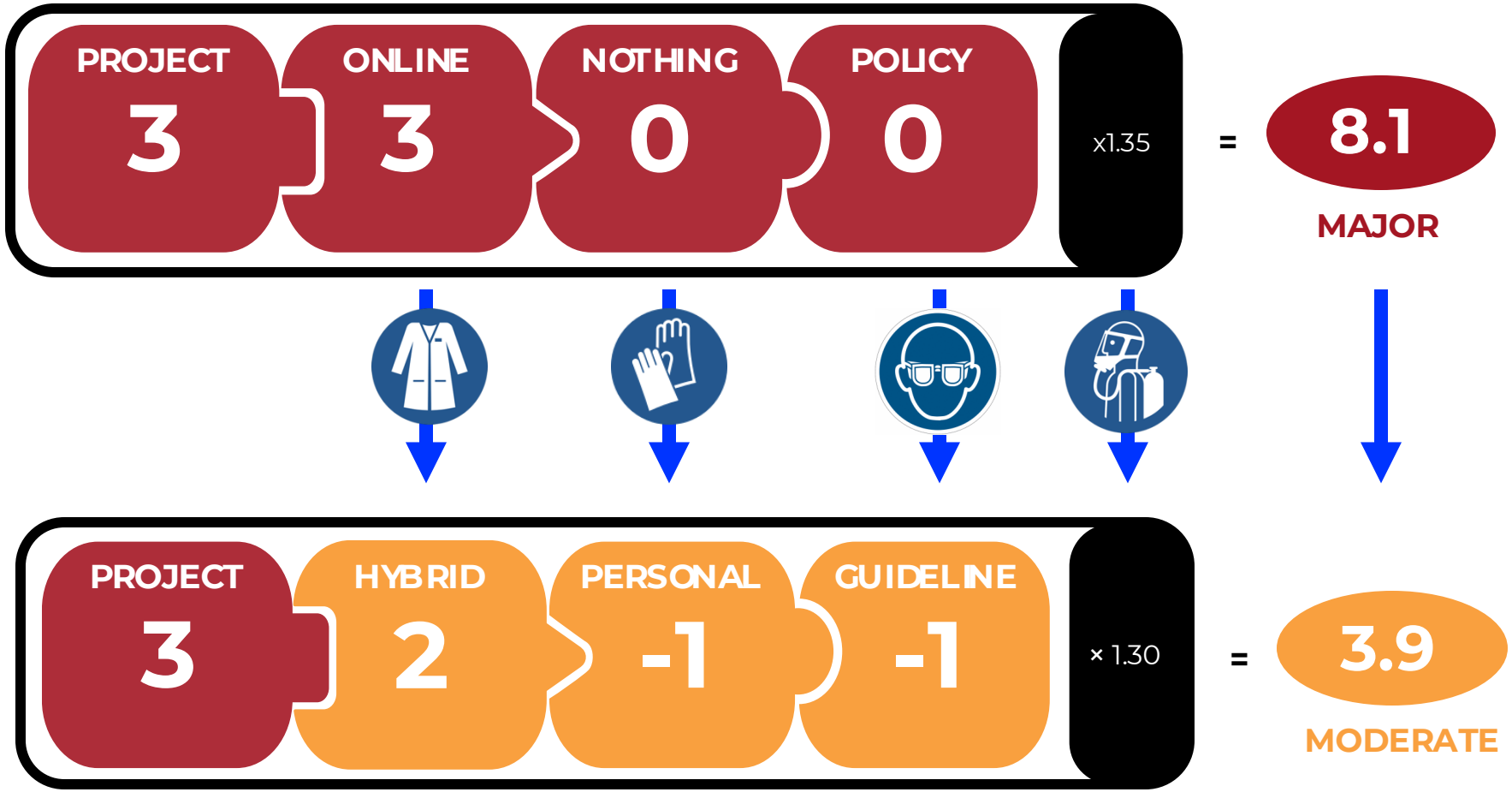

A new shared language

Our dimensional model equips academics with a common vocabulary to discuss AI risk in their assessments

Future directions?

- Refine scoring system
- Consider additional mitigations
- Consider faculty-wide adoption
- Embed risk awareness as *routine* in assessment design

Don’t let GenAI stop you from innovating or doing your assessments

Questions?

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